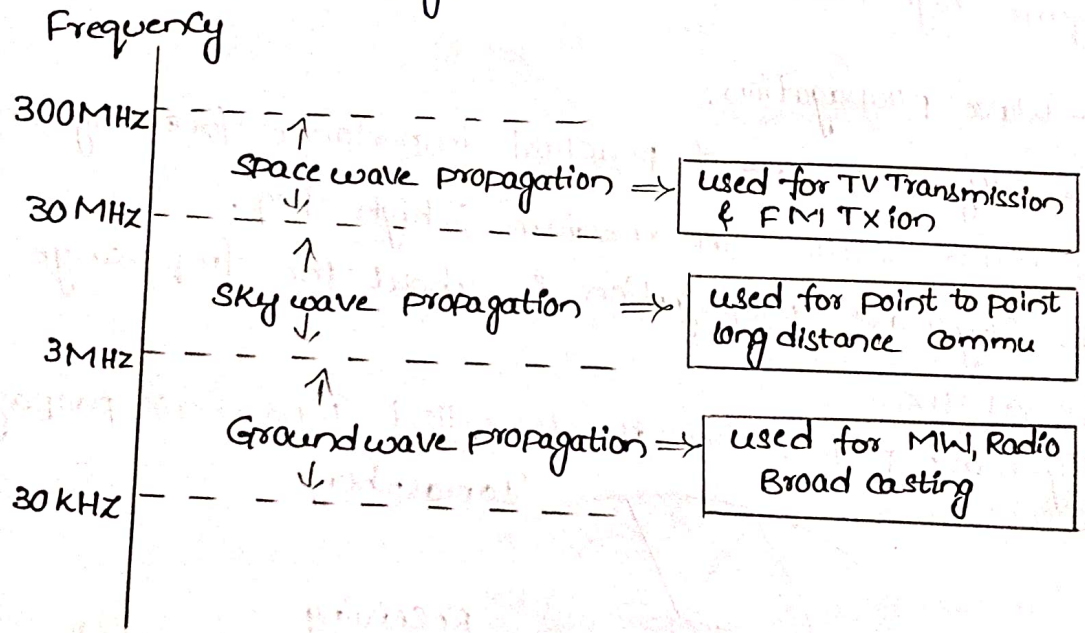
# Module-5
## Wave Propagation

1. Explain the different modes of wave propagation.
2. What are the different paths used for propagating radio waves from 300 KHz to 300 MHz.

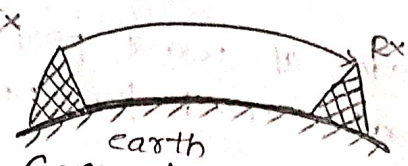
There are 3 different modes of propagation

1) Ground wave propagation
2) Sky wave propagation
3) Space-wave propagation




1. Ground wave propagation:

* The waves which are propagated near the earth's surface are called "Ground waves".
* The frequency range of ground wave propagation is 300KHz to 3MHz
* The ground wave propagation is possible when the Transmitting and Receiving antenna both are closed to the earth's surface.
* This type of propagation is used for MW, Radio Broadcasting.

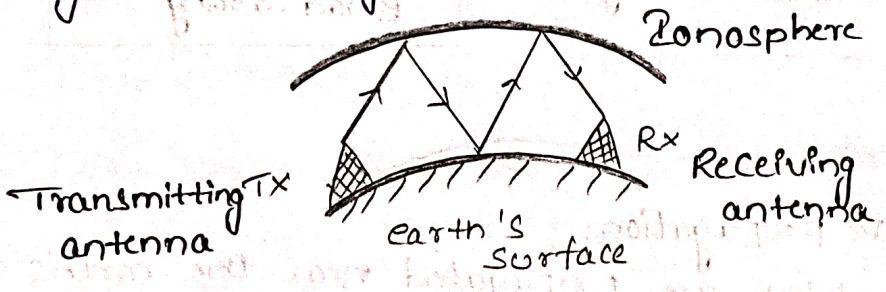


Ground wave propagation

* The ground wave propagation is about LF and MF Frequencies.
* The ground waves are vertically polarized waves, It should require high power for Transmission.

2. Sky-Wave propagation:

* The sky waves are of practical importance for every long radio Communication at medium & high freq.
* The skywave propagation is about the freq range of 3-30 MHz.
* Sky-wave propagation is also called Ionospheric propagation.



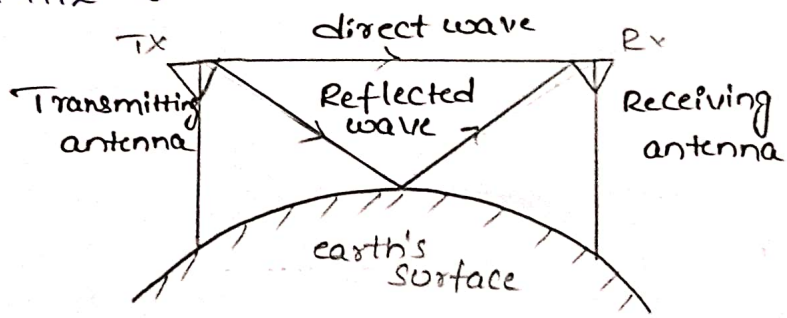

* This mode is used in HF band international broadcasting
* In this mode the EM waves transmitted by the transmitting antenna to reach the receiving antenna.
* EM waves directed in upward at some angle from earth's surface called "sky waves."
* Skywave propagation is used for long distance Commu

3 Space wave propagation:

The radio waves which are having high frequencies called as space waves.

Space waves are the Combination of direct waves & Reflected waves.

Frequency range of space wave propagation is about 30MHz to 300MHz (or) above 30MHz.




* The space wave propagation is composed of direct wave propagation and reflected wave propagation.
* The space wave propagation is through troposphere, hence such propagation is limited to few hundreds of kilometers.
* The spacewave propagation propagates through the freq Bands of HF & VHF freq Bands.

3. Explain about ground wave propagation.

A. **Ground wave propagation** :- Ground wave propogation is a type of radio wave propogation in which the electromagnetic waves travel along the surface of the earth.

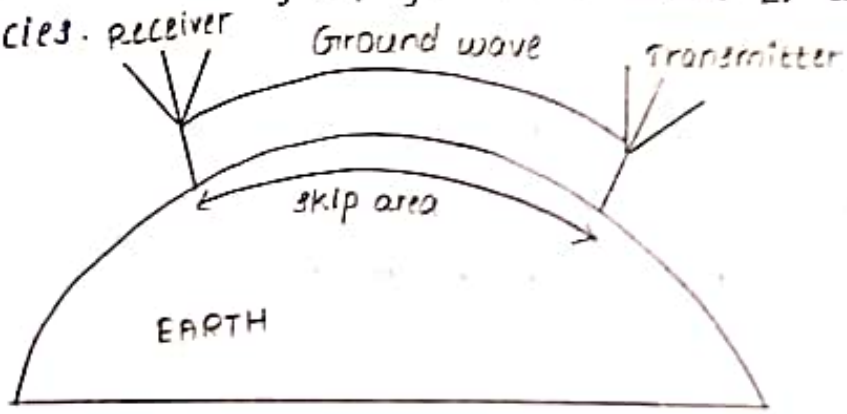
→ The wave follows the curvature of the earth

→ It is mainly used for low and medium frequency signals below 3MHz.

→ The earth's surface and atmosphere guide the wave

→ The wave loses strength due to ground resistance & surface irregularities.

→ The ground wave propagation is about LF and MF frequencies.




→ Hence, the ground wave travels away from the transmitting antenna, it gets attenuated.

$$E = \frac{120\pi h_t \cdot h_r \cdot I_s}{\lambda d} \text{ V/m}$$

where,

* $120\pi = 377\Omega$ = Intrinsic impedance of free space
* $h_t$ and $h_r$ = Effective heights of the transmitting and receiving antennas.

* $I_s$ = Antenna current
* $\lambda$ = wave length
* d = Distance at a point from the transmitter.

**Advantages :-**

→ can cover long distances without needing repeaters.
→ Reliable for local and regional communication.
→ no need for line-of-sight between transmitter and receiver.

**Disadvantages :-**

→ High signal loss due to ground resistance.
→ Requires high transmitting power for long-range communication.

**Applications :-**

→ used in AM radio broadcasting.
→ used in navigation.
→ used in marine communication.

4 Write a short note on 'Troposphere' propagation.

A **Troposphere propagation** :- It is the nearest region of the atmosphere to the earth's surface about 10 to 20km above the earth surface.

→ Troposphere propagation refers to the transmission of radio waves through the lowest layer of the atmosphere.

→ Troposphere is also called as "Region of change".

→ In this type of propagation, radio waves travelled by refraction and scattering due to variations in temperature, pressure and humidity in the troposphere.

→ These variations cause the refractive index to change, bending the waves back towards the earth's surface and allowing communication beyond the line of sight.

**Frequency range** :- Mainly used for VHF (30-300MHz) and UHF (300MHz - 3GHz) signals.

fig(a) :-

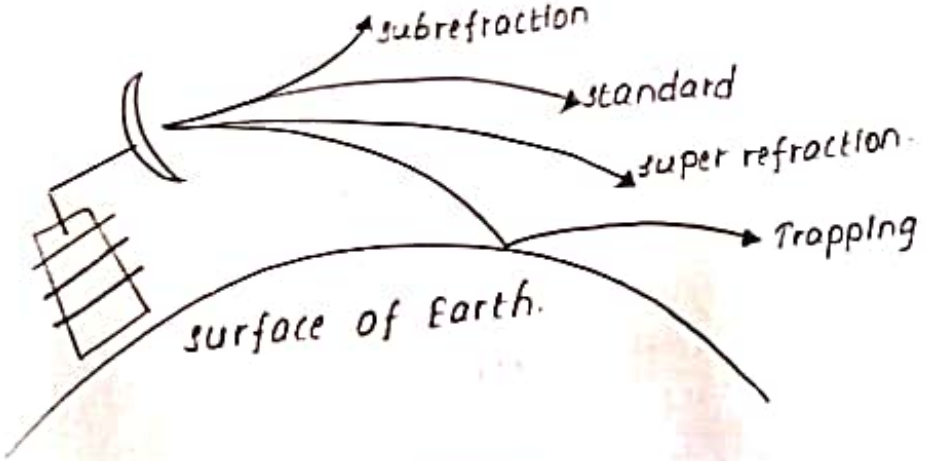
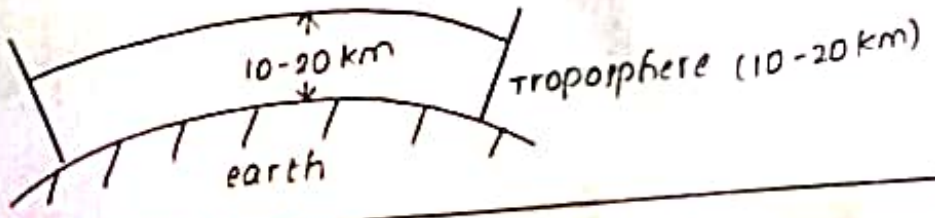



fig (b) :-



Advantages :-

→ supports long - distance communication beyond the horizon.

→ stable and reliable for line - of - sight and slightly beyond.

Disadvantages :-

→ Affected by weather conditions such as rain, fog and temperature changes.

→ Not suitable for long - distance communications.

Applications :-

→ used for TV and FM radio broadcasting.

→ used for microwave links.

→ used for Radar systems.

5. What are the types of propagation and give the typical frequency range of each propagation

A. There are three main modes of propagation

1. **Ground wave propagation:**

Frequency Range : Up to 2 MHz

**Explanation:**

* In this type, the radio wave travel along the surface of the Earth.
* These waves follow the curvature of the Earth due to diffraction.
* The signal strength decreases with decrease with distance of energy loss to the ground.

**Main Characteristics:**

i, Effective for LF bands ~~(0 - 30MHz)~~ ($\leq$ 2 MHz)

ii, Enables long-distances or international communication.

iii, The range increases when transmitting over sea water.

2. **Sky wave propagation:**

Frequency Range : 2 MHz - 30 MHz

**Explanation:**

* In this mode, the transmitted radio waves are directed towards the sky and are reflected or refracted back to Earth by the inosphere.
* sky wave propagation is used for long-distance communication without the need for satellite.
* This reflection allows communication beyond the line of sight, even thousands of kilometers away.

Main characteristics:

i) Effective for High Frequency (HF) bands (2 - 30 MHz)

ii, Enables long-distance or international Communication.

3. Space wave propagation

Frequency Range : Above 30 MHz

Explanation:

* In this mode, the radio waves travel directly from the transmitting Antenna to the Receiving Antenna through the troposphere

* This mode is also known as line-of-sight (LOS)

* The distance covered depends on the height of the Antennas and Earth's Curvature

* space waves are mainly used for UHF, VHF, and microwave frequencies

Main characteristics:

i, Suitable for very high frequency (VHF) and above

ii, Requires directly visibility between transmitter and receiver

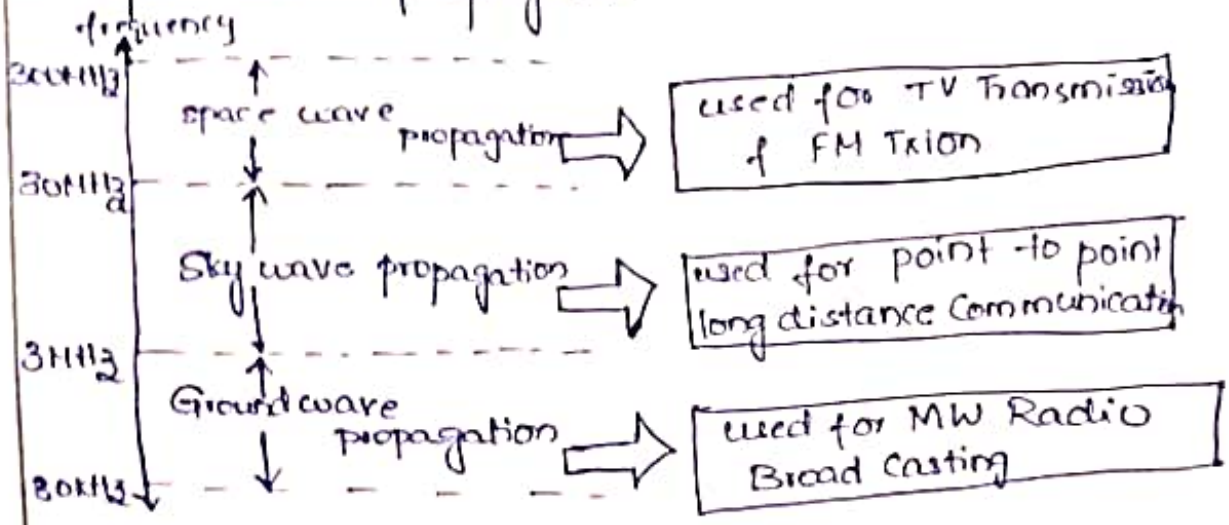
iii, Used for short to medium range high-quality Communication.

iv, Affected by Ostracles like buildings and mountains.

6) Write the Applications of different Modes of propagation?

There are 3 different modes of propagation

1. Ground wave propagation
2. sky wave propagation
3. Space wave propagation




Applications of Different Modes of propagation:

1. Ground Wave propagation:

i) AM Broadcasting in the medium and long-wave bands

ii) Navigation and maritime Communication (e.g., LORAN, NAVTEX)

iii) Military Communication at low frequencies

iv) Emergency Communication Systems.

2. Sky wave propagation:

i) Shortwave Radio Broadcasting.

ii) Amateur (HAM) Radio Communication

iii) International Broadcasting services

iv) Ship-to-shore and Aircraft communication.

v) Defense and military communication

3. Space wave propagation:-

i) FM radio and television broadcasting

ii) Radar Communication

iii) Microwave and satellite Communication links

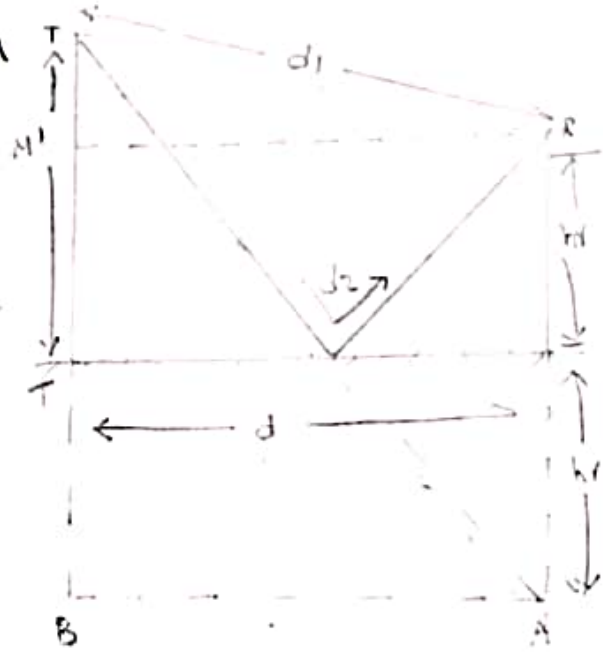
iv) Mobile and cellular Communication systems.

Summary Table:

| Mode | Frequency Range | Applications |
|---|---|---|
| Ground wave | Up to 2 MHz | AM Broadcasting, navigation, marine-communication |
| sky wave | 2-30 MHz | Shortwave, International HAM Radio |
| space wave | Above 30 MHz | FM, TV, radar, satellite, mobile systems. |

Q7) Derive the expression for the electric field strength at the receiving point due to the space wave propagation

A) If the curvature of earth is neglected the space wave propagation takes place as illustrated




let $h_t$ = height of transmitting antenna

$h_r$ = height of recieving antenna

d = distance b/w tem

$d_1$ = direct ray's path.

$d_2$ = indirect ray's path.

$E_0$ = Field strength at R' due to direct rays

From $\Delta T'R'N'$

$$(h_t - h_r)^2 + d^2 = d_1^2$$

//y $\Delta$ T'AB

$$(h_t + h_r) + d^2 = d_2^2$$

$$d_1 = [d^2 + (h_t - h_r)^2]^{1/2}$$

$$= d\left[1 + \left(\frac{h_t - h_r}{2}\right)^2\right]^{1/2} = d\left[1 + \frac{1}{2}\left[\frac{h_t - h_r}{d}\right]^2 + \dots\right]$$

$$d_1 = d + \left(\frac{h_t + h_r}{2d}\right)^2$$

//y

$$d_2 = d + \left(\frac{(h_t + h_r)^2}{2d}\right)$$

Path difference

$$P.d = d_2 - d_1$$

$$= d + \left(\frac{(h_t + h_r)^2}{2d}\right) - d = \frac{(h_t - h_r)^2}{2d} = \frac{2h_t h_r}{2d} + \frac{2h_t h_r}{2d} = \frac{4h_t h_r}{2d}$$

$$P.d = \frac{2h_t h_r}{d}.$$

$$\text{Phase diff}(\alpha) = \frac{2\pi}{\lambda}(\text{Path diff}) = \frac{2\pi}{\lambda}\left(\frac{2h_t h_r}{d}\right)$$

$$\alpha = \frac{4\pi h_t h_r}{d\lambda} \text{ radians}$$

$\theta = \alpha + \beta$

$\alpha$ = phase diff path differences

$\beta$ = phase diff due to reflection.

$$E_R = E_0[1 + ke^{-j\theta}].$$

$$E_R = E_0(1 + k(\cos\theta - j\sin\theta))$$

$$|E_R| = E_0\sqrt{(1 + k\cos\theta)^2 - (jk\sin\theta)^2}$$

$$= E_0\sqrt{1 + k^2\cos^2\theta + 2k\cos\theta + k^2\sin^2\theta} \Rightarrow E_0\sqrt{1 + k^2 + 2k\cos\theta}$$

$$|E_R| = E_0\sqrt{1 + 1^2 + 2\cdot 1\left(2\cos^2\frac{\theta}{2} - 1\right)}$$

$$= E_0\sqrt{2 + 4\cos^2\frac{\theta}{2} - 2} = E_0 2\cos\frac{\theta}{2}$$

$$= 2E_0\cos\left[\frac{\alpha + \pi}{2}\right] = 2E_0\sin\frac{\alpha}{2}$$

$$|E_R| = 2E_0\sin\left[\frac{4\pi h_t h_r}{2d\lambda}\right]$$

$$\sin\frac{4\pi h_t h_r}{2d\lambda} \equiv \frac{2\pi h_t h_r}{2d\lambda}$$

$$|E_R| = 2E_0 \cdot \frac{4\pi h_t h_r}{2d\lambda} = \frac{E_0 4\pi h_t h_r}{d\lambda}$$

$$E_0 = \frac{E_f}{d}$$

$E_f = 7\sqrt{P}$ volt/metre

$P$ = Effective power radiated.

$$E_0 = \frac{7\sqrt{P}}{d}.$$

$$|E_R| = \frac{7\sqrt{P}}{d} \cdot \frac{4\pi h_t h_r}{d\lambda} = \frac{7 \times 4 \times 3.14\sqrt{P} \cdot h_r}{d^2\lambda} = \frac{87.92\sqrt{P}\, h_t h_r}{d^2\lambda}$$

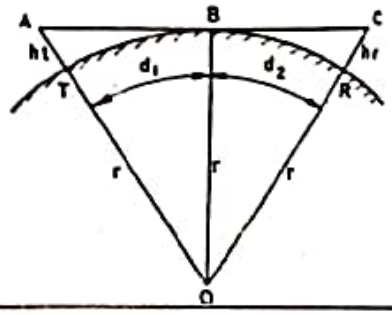
$$|E_R| = \frac{88\sqrt{P}\, h_t h_r}{\lambda d^2} \text{ volt/metre}$$

$$E_R \propto \sqrt{P}$$
$$\propto h_t$$
$$\propto h_r$$
$$\propto \frac{1}{\lambda}$$
$$\propto \frac{1}{d^2}.$$

//.

**9) Derive the expression for LOS range for space wave propagation and explain super refraction?**

**Range of space wave propagation/line of sight distance:**




Let $d$ be the distance between transmitter and the receiver, and heights of the transmitting and receiving antennas are $h_t$ and $h_r$ respectively above ground. Now from Fig. 11.33, the line of sight distance

$$d = d_1 + d_2 \qquad \text{... (11.104)}$$

If $r$ be the radius of earth (equal to 6370 km) then from $\Delta ABO$ and $\Delta CBO$,

$$d_1 = \sqrt{(h_t + r)^2 - r^2} = \sqrt{h_t^2 + r^2 + 2h_t \cdot r - r^2} \simeq \sqrt{2rh_t} \text{ metres} \quad \Big| \because h_t^2 << 2rh_t$$

Similarly, $d_2 = \sqrt{(h_r + r)^2 - r^2} = \sqrt{h_r^2 + r^2 + 2h_r \cdot r - r^2} = \sqrt{2r \cdot h_r}$ metres $\quad \because hr^2 \leq 2rh_r$

Thus putting the values of $d_1$ and $d_2$ in eqn. 11.104, we get

$$d = [\sqrt{2r \cdot h_t} + \sqrt{2rh_r}] \text{ metres} = \sqrt{2r}\,[\sqrt{h_r} + \sqrt{h_r}] \text{ metres} \qquad \text{... (11.104 a)}$$

$$= \sqrt{2 \times 6370 \times 10^3}\,[\sqrt{h_t} + \sqrt{h_r}] \text{ metres} \quad \Big| \because r = 6370 \text{ km} = 6370 \times 10^3 \text{ metres}$$

$$= \sqrt{12.74 \times 10^6}\,[\sqrt{h_t} + \sqrt{h_r}] \text{ metres}$$

$$= 10^3 \sqrt{12.74}\,[\sqrt{h_t} + \sqrt{h_r}] \text{ metres} = 10^3 \times 3.570\,[\sqrt{h_t} + \sqrt{h_r}] \text{ metres}$$

$$\boxed{d = 3.57\,[\sqrt{h_t} + \sqrt{h_r}] \text{ km}} \qquad \text{... (11.105)}$$

Where $h_t$, $h_r$ are heights of transmitting and receiving antennas in metres and the maximum line of sight distance covered by space wave propagation $d$ is in km.

**Super refraction:**

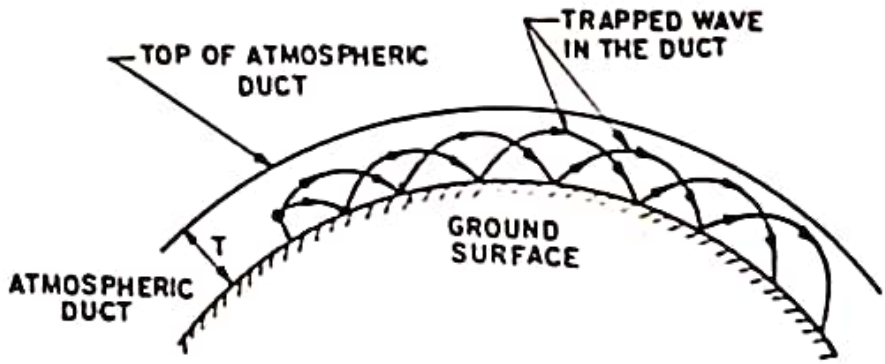



*Fig. 11.37. Super-refraction in the atmospheric duct.*

Where the refracting conditions are sufficiently different from the *standard* to cause trapping of the wave, *the concept of an effective earth radius does not hold good*. In order to give the necessary curvature the actual refractive index $\mu$ at a height $h$ must be replaced by a *modified refractive index* given by

$$N = \mu + \frac{h}{r}$$ | Appendix 11.1

Although $N$ is always close to unity, yet its actual value is important and hence when dealing with numerical values, it is convenient to introduce the excess *modified index of refractive modulus* $M$, related to $N$ as

$$N - 1 = \left(\mu - 1 + \frac{h}{r}\right)$$

or

$$(N - 1) \times 10^6 = \left(\mu - 1 + \frac{h}{r}\right) \times 10^6$$

or

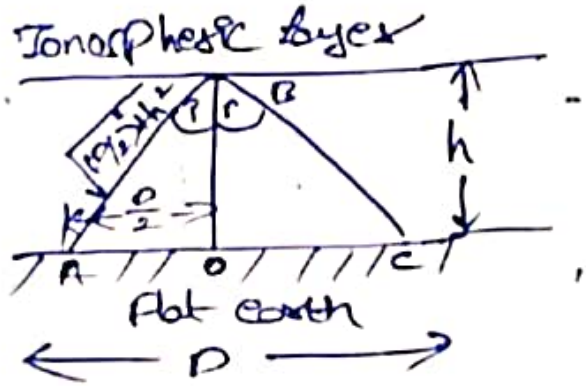
$$\boxed{M = (N - 1) \times 10^6 = \left(\mu - 1 + \frac{h}{r}\right) \times 10^6} \quad \text{...(11.132)}$$

Where $\mu$ = Refractive index ; $h$ = height above ground

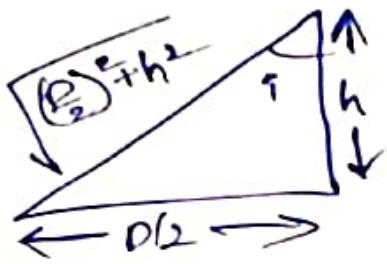
$r$ = True radius of earth = 6370 km.

The value of gradient *i.e.* $\frac{dM}{dh}$ and its sign both depends on the tropospheric conditions. From the measurement, if $M$ is plotted against height $h$, the following different curves are obtained.

**10) Derive the expression for MUF. Explain the terms of ' virtual height ' and ' skip distance ' ?**




$$\cos\theta = \frac{OB}{AB} = \frac{h}{\sqrt{\left(\frac{D}{2}\right)^2 + h^2}}$$




$$= \frac{h}{\sqrt{\frac{D^2}{4} + h^2}}$$

$$= \frac{2h}{\sqrt{D^2 + 4h^2}}$$

$$\cos\theta = \frac{2h}{\sqrt{D^2 + 4h^2}} \quad \text{—(1)}$$

Here, $f = f_{muf}$

$R = 90°$

$M = N_{max}$

$$\mu = \frac{\sin i}{\sin r} = \sqrt{1 - \frac{81 N_{max}}{f_{muf}^2}}$$

$$\sin i = \sqrt{1 - \frac{f_c^2}{f_{muf}^2}}$$

$$f_c = 9\sqrt{N_{max}}$$

$$f_c^2 = 81 N_{max}$$

S.O.B.S

$$\sin^2 i = 1 - \frac{f_c^2}{f_{muf}^2}$$

$$\frac{f_c^2}{f_{muf}^2} = 1 - \sin^2 i$$

$$\cos^2 i = \frac{f_c^2}{f_{muf}^2} \quad \text{——— (2)}$$

$$\cos i = \frac{f_c}{f_{muf}}$$

sub (1) in (2)

$$\cos^2 i = \frac{f_c^2}{f_{muf}^2} = \frac{4h^2}{D^2 + 4h^2}$$

$$\frac{f_{muf}^2}{f_c^2} = \frac{D^2 + 4h^2}{4h^2}$$

$$\frac{f_{muf}}{f_c} = \sqrt{\frac{D^2 + 4h^2}{4h^2}}$$

$$\frac{f_{muf}}{f_c} = \sqrt{1 + \frac{D^2}{4h^2}}$$

$$\boxed{f_{muf} = f_c \cdot \sqrt{1 + \left(\frac{D}{2h}\right)^2}}$$

where, $f_c$ = critical freq

$D$ = skip distance

$h$ = virtual height

$f_{muf}$ = Max usable freq

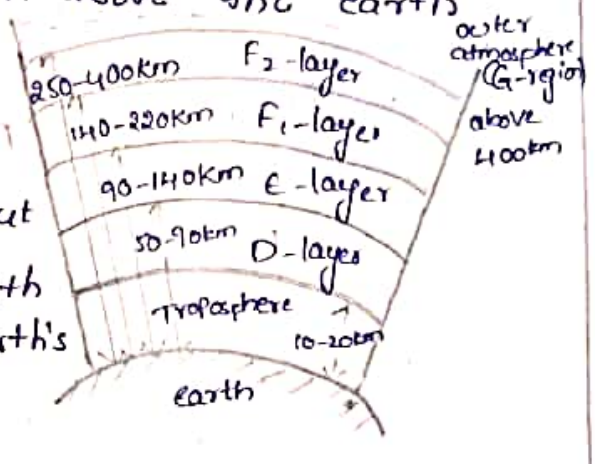
11. Explain the structure of ionosphere and derive the relation between MUF and skip distance.

(A) Structure of Ionosphere:-

→ Ionosphere is the upper portion of the atmosphere of the earth.

→ It gets heated due to the large absorption of large energy radiated by the sun. After heating it get Ionized

→ this Region is located about 70km above the earth surface and upto 400 km.




D-layer:- The D-layer is located about 50 to 90 km above the surface of earth and it is nearest layer to the earth's surface

* It's thickness is about 10km.
* The critical frequency is about 100KHz D-layer present at Day time only.

E-layer:- The E-layer is located about 90-140km above the earth surface.

* thickness is about 25 km.
* critical frequency is about 3 to 5 MHz It provides better Deception during night time.

$F_1$, $F_2$ layer:- The F-layer is located at the height of 140 to 60 km and it is mainly combination of $F_1$ layer (140-220km) and $F_2$ layer (250 to 400 km). During night $F_1$-layer combines with $F_2$-layer and at height of 140-300km, we get F layer.

* critical frequency is 5 to 12 MHz.
* The F1 layer reflects the high frequency waves.
* The F2 layer reflects the high frequency radio waves.

The relation b/w MUF & Skip distance

For a given frequency $f = f_{MUF}$, The skip distance can be calculated as follows.

$$F_{muf} = f_{cr}\sqrt{1+\left(\frac{D_{skip}}{2h}\right)^2}$$

$$\Rightarrow \frac{F_{muf}}{f_{cr}} = \sqrt{1+\left(\frac{D_{skip}}{2h}\right)^2}$$

$$\Rightarrow \left(\frac{F_{muf}}{f_{cr}}\right)^2 = 1+\left(\frac{D_{skip}}{2h}\right)^2$$

$$\Rightarrow \left(\frac{f_{muf}}{f_{cr}}\right)^2 - 1 = \left(\frac{D_{skip}}{2h}\right)^2$$

$$\therefore \frac{D_{skip}}{2h} = \sqrt{\left(\frac{f_{muf}}{f_{cr}}\right)^2 - 1}$$

$$\boxed{D_{skip} = 2h\sqrt{\left(\frac{f_{muf}}{f_{cr}}\right)^2 - 1}}$$ //

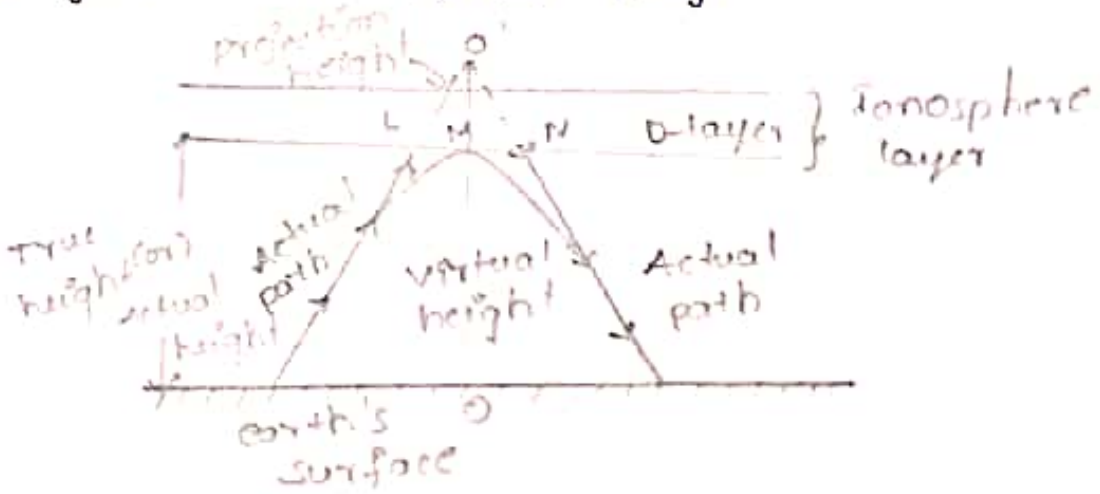
(12) Explain absorption, virtual height and skip distance

Absorption :-

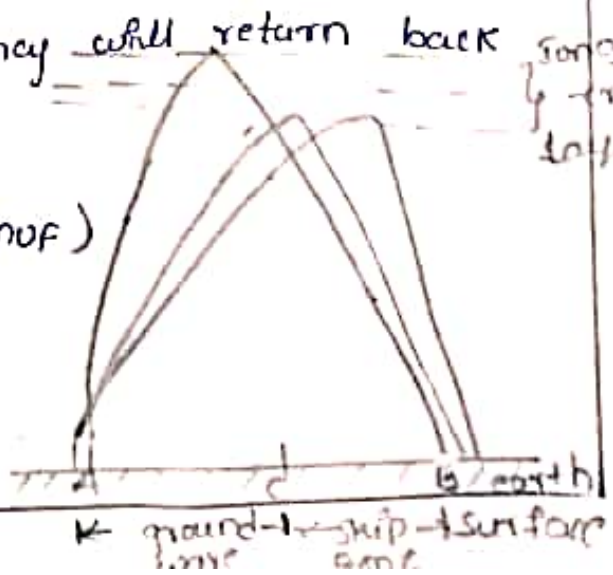
Absorption is the process by which one substance takes in or soaks up another substance. It can occur in different contexts - physical, chemical, or biological.

Virtual height :-

It is defined as the height to which, a short pulse of energy transmitt along vertically upwards and a wave travelling with the speed of light, the virtual height is greater than actual height.




Skip distance :- ($d_{skip}$)

The skip distance is the shortest distance from from the transmitter, measured along surface of the earth, at which a sky-wave of fixed frequency will return back from Ionosphere to earth.

(for a given frequency $f = f_{muf}$)